

# A Geometric-Null Binomial Framework for Harmonic Longitude Proximity Testing in Monument Distributions

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## Highlights

- Geometric-null binomial is the test matched to harmonic-proximity hypotheses.
- Rayleigh concentration and binomial proximity enrichment can dissociate.
- Dissociation demonstrated on the UNESCO dome/stupa corpus ( $N = 90$ ).
- Worked example: Wiki+OSM Fisher  $\chi^2(4) = 28.74$ ,  $p < 0.001$ , Tier A (dome-excluded).
- Max-statistic anchor-sweep correction yields marginal  $p = 0.0753$ .

## Abstract

This paper addresses a test–hypothesis mismatch in spatial periodicity analysis: the Rayleigh statistic measures mean phase concentration at period  $T$ , whereas a repeated-line grid hypothesis predicts thresholded proximity to any harmonic longitude — two summaries that can dissociate within the same corpus. The methods contribution is the geometric-null binomial framework: (i) define phases  $\theta_i = 2\pi((\lambda_i - \lambda_0) \bmod T)/T$ ; (ii) compute harmonic deviation  $\delta_i = \min(r_i, T - r_i)$ ; (iii) test the count of sites with  $\delta_i \leq d$  against the geometric null  $p_0 = 2d/T$ ;

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(iv) combine across independently assembled corpora via Fisher’s method; and  
(v) apply a max-statistic anchor-sweep correction where the anchor requires validation. A period sweep of the UNESCO dome/stupa corpus ( $N = 90$ ) demonstrates the Rayleigh–proximity dissociation: Rayleigh is significant at three periods lacking proximity enrichment, while the  $3^\circ$  period shows proximity enrichment with non-significant Rayleigh. Applied to three hemispherical-monument corpora at the  $3^\circ$  period and  $\sim 35.25^\circ\text{E}$  Levantine corridor phase, the worked example finds consistent corpus-level proximity enrichment. The result is band-invariant: any co-grid longitude within  $\pm 0.06^\circ$  of this phase (including Jerusalem and Mount Gerizim at  $35.269^\circ\text{E}$ , a topographic summit absent from all monument corpora) produces identical proximity counts. The dome-excluded two-corpus combination (Wikidata + OSM) gives  $\chi^2(4) = 28.74$ ,  $p < 0.001$  at Tier A (threshold  $\pm 0.10 \times T$ ). A max-statistic anchor-sweep correction over 36,000 trial longitudes yields an anchor-sweep-corrected  $p = 0.0753$  (marginal), confirming the frame-conditional results cannot be treated as frame-free results without independent anchor motivation. Following [Silva \(2020\)](#), all results are treated as statistical description of spatial structure; questions of mechanism and historical transmission require independent evidence.

**Keywords:** spatial statistics, circular statistics, harmonic proximity, binomial geometric null, monument distribution, archaeoastronomy

## 1 Introduction

Do monuments cluster near specific longitude intervals? The question recurs in the archaeoastronomical and archaeogeographic literature, usually motivated by observational coincidences involving sacred sites, trade corridors, or cosmographic traditions. This paper focuses on *hemispherical monuments* — structures whose principal form is a rounded or mounded mass, including Buddhist stupas, corbelled tholos tombs, and domed shrines — as a corpus for testing longitude-grid hypotheses. This broader question has also been criticized on grounds of investigator-assembled site lists, insufficient correction for multi-

ple comparisons, and lack of independent replication (Williamson & Bellamy, 1983; Freeman, 1976; Kendall, 1974). These concerns motivate a stricter separation between statistical detection and archaeological explanation. Alongside these critiques, this paper raises a complementary issue of test–hypothesis matching: the standard circular-statistical tool used in this literature measures a different quantity than the one a repeated-line grid hypothesis predicts.

The tool in question is the Rayleigh statistic  $R$ . In its circular-statistical form, Rayleigh’s mean-resultant test is a standard test of departure from uniformity toward a single preferred direction on the circle (Rayleigh, 1919; Fisher, 1993; Mardia & Jupp, 2000), most commonly encountered in archaeology and skyscape work in orientation and azimuth analysis (Magli, 2016; Silva, 2020). Here it becomes relevant because longitudes can be folded modulo a candidate period  $T$ :  $R$  then measures whether the folded phases concentrate toward one direction.

This is a useful diagnostic, but only a partial answer to the repeated-line question. A grid-placement hypothesis predicts that sites fall near longitudes  $\lambda_0, \lambda_0 + T, \lambda_0 + 2T, \dots$ . Under modulo- $T$  phase mapping these harmonics collapse to a common phase, so an ideal grid-aligned corpus would produce strong Rayleigh concentration. In real geographically clustered corpora, however, Rayleigh concentration and thresholded harmonic proximity can dissociate: a dominant off-harmonic cluster can produce significant Rayleigh concentration, while a corpus with many near-harmonic sites plus substantial off-harmonic mass can produce binomial enrichment without significant Rayleigh  $R$ . A non-significant Rayleigh result therefore does not rule out harmonic proximity, and a significant Rayleigh result does not by itself establish proximity to repeated longitude lines. The two statistics are complementary; they answer related but distinct questions.

The directly matched test for “sites cluster near harmonics” is straightforward. Under a uniform longitude model, the expected fraction of longitudes falling within  $\pm d^\circ$  of any harmonic of period  $T$  is  $p_0 = 2d/T$ . This is the *geometric null*. In clustered corpora it remains a useful geometric benchmark, but the resulting  $p$ -values are interpreted here as corpus-level proximity enrichment rather than counts of independent placement decisions.

This paper develops that matched test and uses the monument data as a worked methodological example. Under a fixed period, fixed corridor phase, fixed proximity thresholds, and fixed operational corpora, the geometric-null binomial test estimates whether more sites fall near harmonic longitudes than expected from the geometric coverage of the threshold bands. Rayleigh statistics are retained as complementary circular diagnostics, and the period sweep is used to show when phase concentration and thresholded harmonic proximity agree or dissociate. The empirical application illustrates corpus-level proximity enrichment at the fixed  $3^\circ$  period and  $\sim 35.25^\circ\text{E}$  Levantine corridor phase. Following [Silva \(2020\)](#), this is treated throughout as statistical description of spatial structure in an open-data worked example; historical, cultural, or geographic explanation is a separate interpretive task requiring independent evidence.

## 2 Rayleigh Concentration and Harmonic Proximity: Complementary but Distinct Statistics

### 2.1 Why the Two Statistics Can Dissociate

Let  $\lambda_1, \dots, \lambda_N$  be site longitudes and  $T$  a candidate grid period with anchor  $\lambda_0$ . Define phases by  $\theta_i = 2\pi((\lambda_i - \lambda_0) \bmod T)/T$ , and the Rayleigh statistic by

$$R = \left| \frac{1}{N} \sum_{i=1}^N e^{i\theta_i} \right|,$$

where  $i = \sqrt{-1}$  (complex exponential; equivalently,  $e^{i\theta} = \cos \theta + i \sin \theta$ , so  $R$  is the length of the mean of the unit vectors  $(\cos \theta_i, \sin \theta_i)$ ).

A key property of this transformation is that sites near *any* harmonic of  $T$  — near  $\lambda_0$ , near  $\lambda_0 + T$ , near  $\lambda_0 + 2T$ , and so on — all map to the same phase  $\theta \approx 0$  under the modulo- $T$  reduction, because  $(\lambda_0 + kT - \lambda_0) \bmod T = 0$  for any integer  $k$ . Under ideal conditions (all sites falling exactly on a harmonic),  $R$  would equal 1. The modulo- $T$  phase map therefore does not, by itself, produce multiple phase classes: harmonics collapse to one class, not many.

The Rayleigh test and the binomial proximity test can nevertheless dissociate because they aggregate cross-site information differently.

*What Rayleigh measures.*  $R$  is the length of the mean phasor: it is sensitive to whether a *single direction* on the unit circle predominates after averaging over the full corpus. In a geographically clustered corpus, a single dominant geographic cluster can produce large  $R$  at period  $T$  simply because that cluster’s longitude maps to a concentrated phase at that particular  $T$  — regardless of whether the cluster is harmonically proximate. This is a generic property of clustered data on the unit circle; it can produce significant  $R$  at periods that carry no harmonic-proximity interpretation.

*What the binomial measures.* The binomial proximity test counts the number of sites whose deviation from the nearest harmonic falls within a threshold  $d$ :  $\#\{i : \delta_i \leq d\}$ . Let  $r_i = ((\lambda_i - \lambda_0) \bmod T)$  and  $\delta_i = \min(r_i, T - r_i)$ . This is independent of the mean phase direction. Sites far from harmonics (large  $\delta_i$ ) contribute zero to the binomial count regardless of whether they push  $R$  up or down. Conversely, a corpus can show significant binomial enrichment even when many off-harmonic sites keep the mean phasor small.

*Complementarity.* The two tests ask complementary questions. Rayleigh asks: does one phase direction dominate? The binomial asks: are more sites within a fixed reporting threshold of a harmonic than the uniform null predicts? In a real corpus these can agree or disagree, depending on the balance between harmonically proximate sites, off-harmonic sites, and the geographic clustering structure of the corpus. A non-significant Rayleigh result does not rule out harmonic proximity; a significant Rayleigh result does not establish harmonic proximity unless the phase concentration is verified to coincide with a harmonic. Relying on Rayleigh alone to evaluate a repeated-line hypothesis can therefore produce false positives at cluster-dominated periods and miss genuine proximity enrichment when near-harmonic sites coexist with substantial off-harmonic mass.

## 2.2 Empirical Demonstration: Period Sweep Dissociation

The following demonstration uses the UNESCO keyword-matched dome/stupa corpus described fully in Section 4. Table 1 illustrates the dissociation concretely ( $N = 90$ ).

The full sweep of 46 candidate periods ( $1.5^\circ$  to  $6.0^\circ$  in  $0.1^\circ$  steps) is available in the supplementary audit; four representative periods are shown here to illustrate the qualitatively distinct outcomes. Both tests use the Tier-A threshold ( $\pm 0.10 \times T$ , geometric null  $p_0 = 20\%$ ; tiers defined in Section 3.2); finer-threshold tiers (A+, A++) are reported in the primary results.

Table 1: Rayleigh concentration vs. Tier-A binomial proximity enrichment at four representative periods drawn from a full sweep of 46 candidate periods ( $1.5^\circ$  to  $6.0^\circ$  in  $0.1^\circ$  steps; complete sweep tables in supplementary audit). These four periods span the full range of dissociation outcomes: three Rayleigh-significant periods without proximity enrichment, and the one period showing proximity enrichment with non-significant Rayleigh concentration. UNESCO dome corpus ( $N = 90$ , corridor phase  $\sim 35.25^\circ\text{E}$ ). Tier-A threshold  $= 0.10 \times T$ ; geometric null  $p_0 = 20\%$ . Sig symbols:  $*p < 0.05$ ,  $**p < 0.01$ , ns = not significant.

| Period      | Rayleigh $p$ | Sig | Tier-A hits | Expected | Binom. $p$ | Sig |
|-------------|--------------|-----|-------------|----------|------------|-----|
| $2.4^\circ$ | 0.0083       | **  | 21          | 18.0     | 0.2503     | ns  |
| $3.0^\circ$ | 0.1796       | ns  | 29          | 18.0     | 0.0043     | **  |
| $3.5^\circ$ | 0.0267       | *   | 13          | 18.0     | 0.9312     | ns  |
| $3.6^\circ$ | 0.0407       | *   | 20          | 18.0     | 0.3383     | ns  |

The dissociation is clear. Rayleigh is significant at  $2.4^\circ$ ,  $3.5^\circ$ , and  $3.6^\circ$ , and non-significant at  $3.0^\circ$ . The binomial proximity test is significant only at  $3.0^\circ$ . A researcher relying on Rayleigh alone to evaluate the repeated-line hypothesis would flag periods that carry no proximity enrichment, and would overlook the period with the clearest proximity structure. Using the two tests together avoids this.

The pattern is consistent with the mechanism described above. At  $2.4^\circ$ ,  $3.5^\circ$ , and  $3.6^\circ$ , one geographic cluster dominates the phase distribution, producing concentrated  $R$  without harmonic proximity enrichment. At  $3.0^\circ$ , many sites fall near the harmonic class, but substantial off-harmonic mass keeps the mean phasor small; the binomial test correctly registers the thresholded per-site proximity counts. Rayleigh and the binomial measure genuinely different aspects of the data; neither is wrong, but only the binomial directly answers the question “do sites fall near harmonics?”

A broader candidate-period survey of the dome/stupa corpus ( $1.5^\circ$  to  $6.0^\circ$  in  $0.1^\circ$  steps, 46 periods; four representative periods shown in Table 1) finds the dominant binomial proximity peak at  $3^\circ$  ( $p = 0.0043$ ); selected nearby non-harmonic spacings ( $1.8^\circ$ ,  $2.1^\circ$ ,

148  $2.7^\circ$ ,  $3.3^\circ$ ) are non-significant in the same sweep. This motivates  $3^\circ$  as the primary  
 149 period for the cross-corpus analysis in Section 5.

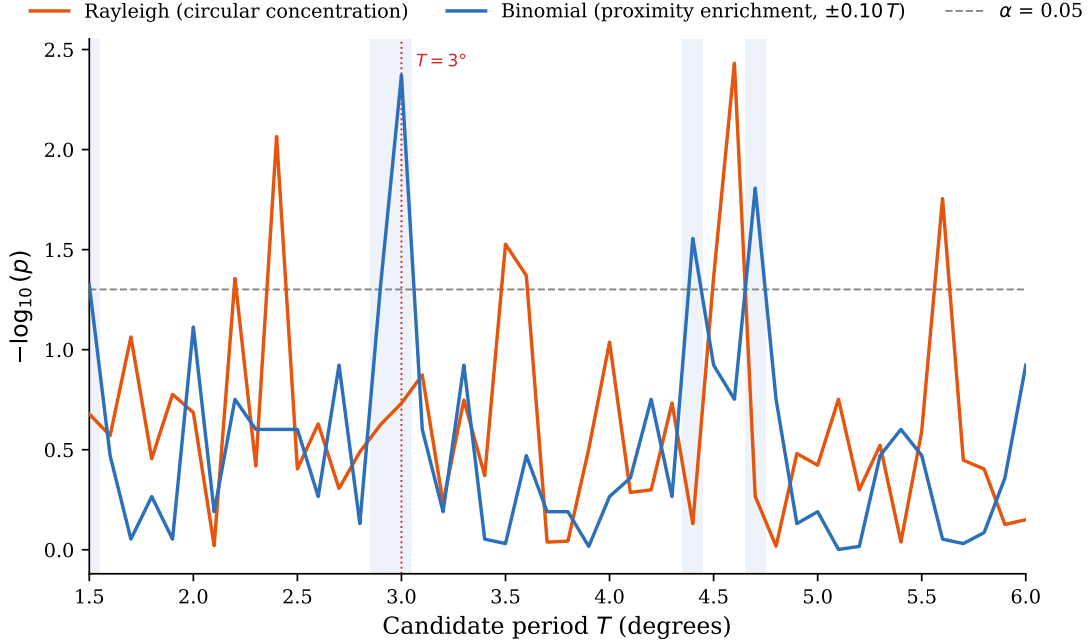


Figure 1: Period sweep of the UNESCO dome/stupa corpus ( $N = 90$ ):  $-\log_{10}(p)$  for the binomial proximity-enrichment test (blue; threshold  $\pm 0.10 \times T$ ) and the permutation Rayleigh test (orange) across 46 candidate periods ( $1.5^\circ$  to  $6.0^\circ$  in  $0.1^\circ$  steps); anchor  $\approx 35.25^\circ\text{E}$ . Dashed line marks  $\alpha = 0.05$ . The blue-shaded column at  $T = 3^\circ$  highlights the dominant proximity peak; Rayleigh is non-significant at this period, illustrating the Rayleigh–binomial dissociation.

150 A Monte Carlo spacing test (100,000 permutations, dome-label shuffle) across 18 can-  
 151 didate spacings ( $0.5^\circ$ – $15^\circ$ ; full list in supplementary audit) finds the observed count of 4  
 152 significant spacings marginal ( $p \approx 0.0979$ ). Three of the four significant spacings ( $6^\circ$ ,  $8^\circ$ ,  
 153  $12^\circ$ ) are multiples of  $3^\circ$  and their hits are fully accounted for by the  $3^\circ$  harmonic structure;  
 154 the  $3^\circ$  spacing is the only one whose significance is not explained by overlap with another  
 155 spacing’s nodes.

## 156 3 The Geometric-Null Binomial Framework

### 157 3.1 The Geometric Null

158 For a grid of period  $T$  anchored at  $\lambda_0$ , define the anchor-relative remainder and nearest-  
 159 harmonic deviation as

$$r_i = ((\lambda_i - \lambda_0) \bmod T), \quad \delta_i = \min(r_i, T - r_i).$$

A site falls within tier threshold  $d$  if  $\delta_i \leq d$ . Under a uniform longitude model, the expected fraction of longitudes satisfying that inequality is  $p_0 = 2d/T$ . In clustered corpora this remains a useful geometric benchmark; the resulting  $p$ -values are interpreted as corpus-level proximity enrichment rather than counts of independent placement decisions, and declustering is not applied because doing so requires an arbitrary choice of scale or linkage criterion with no principled stopping point. The multi-corpus design provides the structural safeguard: the three corpora are dominated by different geographic nodes (the distributed Eurasian corridor for UNESCO, the Java/Sumatra node for Wikidata, and crowd-sourced coverage density across South and Southeast Asia for OSM), so no single cluster can simultaneously account for enrichment across all three independently assembled datasets. Two empirical checks confirm the result is not a generic corridor artifact: the anchor sweep places the Levantine phase at only the 99.7th percentile of all 36,000 trial longitudes (Section 5), and the UNESCO Americas sub-corpus shows no proximity enrichment at the Levantine phase (A+ rate below the geometric null).

The two null models reported in Section 5 (Null A: within-dome bootstrap; Null B: same-footprint UNESCO background, the discriminating comparison) provide complementary assessments of the observed UNESCO proximity enrichment; they are discussed in Section 6.

## 3.2 Tier Classification

We define three proximity thresholds as round fractions of the  $T/2$  half-period ( $1.5^\circ$ ), calibrated to bracket the empirical natural proximity scales of the corpora under study: the Old World Trade Routes (OWTRAD; Ciolek 2004) Silk Road network (2.6%), the UNESCO dome subset (3.6%), the full UNESCO Cultural/Mixed corpus (10.2%), and the Wikidata stupa corpus (20.2%).<sup>1</sup> Each threshold corresponds to a geometric null rate

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<sup>1</sup>These values are the peak-enrichment null rates from the multiscale window sweep (`analysis/unesco/multiscale_combined_p.py`); a standalone reproduction is provided in `supplementary/audit/tier_calibration_audit.py`.

184  $p_0 = 2d/T$  (A++ denotes the finest, strictest threshold; A the most lenient):

|     | Tier | Threshold                                                 | Geometric null $p_0$ |
|-----|------|-----------------------------------------------------------|----------------------|
| 185 | A++  | $\delta \leq 0.06^\circ$ ( $\lesssim 6.66$ km equatorial) | 4%                   |
|     | A+   | $\delta \leq 0.15^\circ$ ( $\lesssim 16.6$ km)            | 10%                  |
|     | A    | $\delta \leq 0.3^\circ$ ( $\lesssim 33$ km)               | 20%                  |

186 Tier labels are descriptive shorthand; the primary test is always the binomial. Because  
187 the thresholds are calibrated to the empirical proximity scales of the corpora under study,  
188 data-adaptive threshold selection of this kind may favor detecting enrichment at the  
189 chosen values; results should be evaluated by the binomial statistics rather than the tier  
190 labels themselves.

### 191 3.3 Cross-Corpus Consistency via Fisher Combination

192 For each corpus  $c$  at tier  $d$ , a one-sided binomial test yields  $p_c$ . Fisher’s combination  
193 statistic  $\chi^2 = -2 \sum_c \ln p_c$ , referred to  $\chi^2(2C)$  for  $C$  corpora, summarizes whether the di-  
194 rectional excess is consistent across all corpora simultaneously. Fisher’s method formally  
195 assumes independent  $p$ -values. The three corpora were assembled by distinct processes  
196 and their geographic clusters are substantially non-overlapping, but they share historical  
197 and geographic structure (Eurasian corridor, stupa tradition, public data sources). The  
198 combined  $p$ -value is therefore used as a cross-corpus consistency summary, noting that  
199 full statistical independence of the three corpora cannot be assumed.

200 The dome corpus period sweep identifies  $3^\circ$  as the dominant proximity peak in that  
201 corpus; this motivates evaluating the independently assembled corpora at the same pe-  
202 riod. Wikidata and OSM were evaluated at  $3^\circ$  without any period sweep of those cor-  
203 pora. This asymmetry is verifiable from the published code and audit files. Following  
204 [Silva \(2020\)](#), all results are treated as statistical description of spatial structure; there is  
205 no prior documentation of a  $3^\circ$  harmonic grid anchored to the Levantine corridor, and  
206 these statistics do not constitute evidence for one absent independent archaeological sup-  
207 port. The two-corpus Fisher restricted to Wikidata and OSM (reported in Table 2 as

“Dome-excl. combined”) summarizes whether the same period shows consistent proximity enrichment across independently assembled datasets.

## 4 Data

**UNESCO dome/stupa subset** ( $N = 90$ ). The UNESCO World Heritage XML dataset (UNESCO World Heritage Centre, 2024) contains 1011 Cultural and Mixed inscribed sites. Seven morphological keywords (stupa/stupas, tholos, dome/domed/domes, spherical) were matched against site names, XML descriptions, and OUV text, yielding  $N = 90$  sites. All statistical tests use this keyword-matched population.

**Wikidata Q180987 stupa corpus** ( $N = 229$ ). All *instance-of: stupa* (Q180987) items in the Wikidata knowledge base, extracted via SPARQL query on the *instance-of: stupa* property. The corpus is geographically dominated by the Java/Sumatra node ( $\sim 105\text{--}115^\circ\text{E}$ ), reflecting the documented historical density of stupa construction around Borobudur (Miksic, 1990).

**OSM stupa layer** ( $N = 259$ ). OpenStreetMap features with tags `building=stupa`, `historic=stupa`, or `ruins=stupa`, assembled by crowd-sourced tagging via Overpass API (OpenStreetMap contributors, 2024). Data © OpenStreetMap contributors, available under the Open Database Licence (ODbL 1.0; <https://opendatacommons.org/licenses/odbl/>). The corpus is treated as a co-primary dataset: it is assembled by a process entirely independent of UNESCO and Wikidata curation, and OSM’s coverage notes are a property of the dataset as a whole rather than a limitation specific to this application.

All three corpora were tested against the same fixed candidate frame: period  $3^\circ$  and the  $\sim 35.25^\circ\text{E}$  Levantine corridor phase. Because open datasets are publicly available, the frame is treated uniformly as fixed for the present analysis across all three corpora. The empirical application therefore functions as a frame-specified spatial test.

**Candidate frame.** The candidate reference phase is the  $\sim 35.25^\circ\text{E}$  Levantine corridor phase, which emerged as the dominant proximity peak phase in the dome corpus

sweep described in Section 2; Wikidata and OSM were evaluated at this phase without independent sweeps of their own. The result is band-invariant: any anchor longitude within  $\pm 0.06^\circ$  of this phase — including Jerusalem at  $35.232^\circ\text{E}$  — produces identical proximity counts, and the data cannot distinguish between co-grid longitudes within the same proximity tier. The anchor is labeled by Mount Gerizim at  $35.269^\circ\text{E}$ , a topographic summit within the Levantine corridor absent from all monument corpora and not derived from any external literature on longitude grids. It serves as a named geographic coordinate for the interval under test; no substantive weight should be attached to the choice of Gerizim over any other co-grid longitude within the same proximity tier.

## 5 Results

### 5.1 Primary Result: Cross-Corpus Geometric-Null Test

Table 2 reports the primary statistical results: one-sided binomial tests and Fisher combinations across all three corpora.

In the worked example, all three corpora exceed the geometric null at every tier despite substantially different geographic footprints and assembly processes. Fisher combinations are reported as cross-corpus consistency summaries (see Section 3 for the independence caveat). The UNESCO A++ result shows the largest individual fold-enrichment ( $3.06\times$ ), reflecting the distributed character of that corpus; Wikidata and OSM A++ results are individually non-significant, reflecting cluster-level rather than individual-site precision. The two-corpus combination restricted to Wikidata and OSM (“Dome-excl. combined”) yields  $\chi^2(4) = 28.74$ ,  $p < 0.001$  at Tier A and  $\chi^2(4) = 12.547$ ,  $p = 0.014$  at Tier A+; the Tier A++ two-corpus combination is non-significant ( $p = 0.346$ ). The all-three-corpus Tier A Fisher combination ( $p < 0.001$ ) includes the dome corpus, which motivated the period, and provides a less conservative cross-corpus summary than the dome-excluded combination. All  $p$ -values are frame-conditional proximity-enrichment measures for the fixed  $3^\circ$  period and  $\sim 35.25^\circ\text{E}$  Levantine corridor phase; following Silva (2020) they are statistical descriptions of spatial structure, not evidence of historical placement. The  $3^\circ$

Table 2: Cross-corpus geometric-null harmonic-proximity enrichment at the fixed  $3^\circ$  period and  $\sim 35.25^\circ\text{E}$  Levantine corridor phase. For each corpus and tier, the one-sided binomial test compares the observed number of sites within distance  $d$  of any harmonic with the expected count under the geometric benchmark  $p_0 = 2d/T$ . The “Dome-excl. combined” row restricts Fisher’s combination to Wikidata and OSM (excluding the period-motivating dome corpus) and is the primary cross-corpus consistency summary; see Section 3 for the independence caveat.

| Tier              | Corpus                                                                              | $N$ | Hits | Exp  | Fold         | $p_{\text{binom}}$ | Sig |
|-------------------|-------------------------------------------------------------------------------------|-----|------|------|--------------|--------------------|-----|
| A++ ( $p_0=4\%$ ) | UNESCO domes                                                                        | 90  | 11   | 3.6  | $3.06\times$ | 0.0009             | *** |
|                   | Wikidata stupas                                                                     | 229 | 11   | 9.2  | $1.2\times$  | 0.311              | ns  |
|                   | OSM stupas                                                                          | 259 | 12   | 10.4 | $1.16\times$ | 0.3433             | ns  |
|                   | <i>All-3 Fisher</i> $\chi^2(6) = 18.402$ , <i>nominal</i> $p = 0.0053$              |     |      |      |              |                    |     |
|                   | <i>Dome-excl. combined (Wiki+OSM)</i> $\chi^2(4) = 4.474$ , $p = 0.346$ , <i>ns</i> |     |      |      |              |                    |     |
| A+ ( $p_0=10\%$ ) | UNESCO domes                                                                        | 90  | 18   | 9.0  | $2.0\times$  | 0.0032             | **  |
|                   | Wikidata stupas                                                                     | 229 | 33   | 22.9 | $1.44\times$ | 0.0212             | *   |
|                   | OSM stupas                                                                          | 259 | 33   | 25.9 | $1.27\times$ | 0.089              |     |
|                   | <i>All-3 Fisher</i> $\chi^2(6) = 24.017$ , <i>nominal</i> $p = <0.001$              |     |      |      |              |                    |     |
|                   | <i>Dome-excl. combined (Wiki+OSM)</i> $\chi^2(4) = 12.547$ , $p = 0.014$ , *        |     |      |      |              |                    |     |
| A ( $p_0=20\%$ )  | UNESCO domes                                                                        | 90  | 29   | 18.0 | $1.61\times$ | 0.0043             | **  |
|                   | Wikidata stupas                                                                     | 229 | 70   | 45.8 | $1.53\times$ | 0.0001             | *** |
|                   | OSM stupas                                                                          | 259 | 69   | 51.8 | $1.33\times$ | 0.0059             | **  |
|                   | <i>All-3 Fisher</i> $\chi^2(6) = 39.661$ , <i>nominal</i> $p < 0.001$               |     |      |      |              |                    |     |
|                   | <i>Dome-excl. combined (Wiki+OSM)</i> $\chi^2(4) = 28.74$ , $p < 0.001$ , ***       |     |      |      |              |                    |     |

period emerged as the dominant proximity peak in the dome corpus sweep (Section 2); the three-corpus cross-corpus analysis is the methodological demonstration that the same statistical pattern appears consistently across independently assembled datasets at this period. A max-statistic anchor-sweep correction (10,000 permutations over 36,000 trial anchors) yields an anchor-sweep-corrected  $p = 0.0753$  ( $z = 1.65$ , marginal), confirming that the frame-conditional results should not be treated as frame-free results; full details are in Section 6.

## 5.2 Supplementary Rayleigh Phase Concentration

Rayleigh statistics are retained as complementary diagnostics for comparability with prior literature. Table 3 reports them at  $T = 3^\circ$ . The full UNESCO corpus shows no Rayleigh phase concentration ( $R = 0.0224$ , perm- $p = 0.6008$ , ns) at  $T = 3^\circ$ , consistent with the proximity pattern not being a generic circular-statistical property of the full heritage list.

Watson’s  $U^2$  and Kuiper tests are retained as corroborating two-sample diagnostics (not independent tests): both show that the dome/stupa phase distribution differs detectably from the full UNESCO corpus ( $p = 0.0036$ ,  $p = 0.0013$  respectively). Because any geographically concentrated subset will tend to differ from its parent corpus in phase distribution, these results confirm concentration rather than proximity enrichment per se; they are included for comparability with prior literature that uses two-sample circular diagnostics as subset-specificity checks.

Table 3: Supplementary Rayleigh phase concentration at  $T = 3^\circ$ , retained for comparability with prior literature. The binomial proximity result (Table 2) is the matched test; Rayleigh statistics are reported here as supplementary diagnostics. All results are conditional on the fixed corridor-phase frame.

| Corpus                                                                                   | $N$  | $R$    | perm- $p$ | Sig |
|------------------------------------------------------------------------------------------|------|--------|-----------|-----|
| UNESCO full corpus                                                                       | 1011 | 0.0224 | 0.6008    | ns  |
| UNESCO dome/stupa                                                                        | 90   | 0.1365 | 0.1883    | ns  |
| Wikidata Q180987                                                                         | 229  | 0.1669 | 0.0016    | **  |
| OSM stupa layer                                                                          | 259  | 0.1073 | 0.0355    | *   |
| <i>Fisher (Wikidata + OSM): <math>\chi^2(4) = 19.551</math>, <math>p = 0.0006</math></i> |      |        |           |     |

### 5.3 Negative Controls and Scope

The OWTRAD Silk Road route network ( $N = 1674$  nodes) does not independently recover the monument phase in the Normalized Phasor Consensus (NPC) test ( $K = |\bar{u}_A \cdot \bar{u}_B|$ , where  $\bar{u}_c$  is the normalized mean phasor of corpus  $c$ ;  $K = 1$  indicates identical phase direction,  $K = 0$  orthogonal; full derivation in Section 5.4;  $K = 0.7429$ , perm- $p = 0.46532$ , ns). Using the 1946 edge midpoints as the proximity sample (one value per route segment, non-duplicated) is the most conservative representation of the network; the binomial test is non-significant at both tiers (A++:  $p = 0.0908$ , ns; A+:  $p = 0.0679$ , ns). The route network therefore shows no enrichment under either the directional or the per-segment proximity test when each segment is counted once.

The UNESCO Americas sub-corpus ( $N = 145$ ) shows A+ rate of 9.7%, below the 10% geometric null, indicating no proximity enrichment at the  $\sim 35.25^\circ\text{E}$  Levantine corridor phase in the Western Hemisphere.

A type-matched geographic control uses the North American mound corpus ( $N = 236$ ): National Historic Landmark (NHL) and National Register of Historic Places (NRHP)-registered earthen burial and platform mounds assembled from the NPS National Register database with coordinates resolved via Wikidata (NPS, 2024; Wikimedia Foundation, 2024). These are broadly mounded earthen structures used here as a geographically distinct type control, located entirely in the Western Hemisphere outside any connection to the  $\sim 35.25^\circ\text{E}$  Levantine corridor phase. At the  $3^\circ$  period and  $\sim 35.25^\circ\text{E}$  Levantine corridor phase the mounds corpus shows no proximity enrichment at any tier: A++: 1/236 (0.4%, null 4%,  $p = 1.00$ , ns); A+: 11/236 (4.7%, null 10%,  $p = 1.00$ , ns); A: 45/236 (19.1%, null 20%,  $p = 0.66$ , ns). All three tiers are at or below the geometric null. This supports the methodological illustration that the Eurasian pattern is geographically specific to the Eurasian corridor rather than a generic outcome of applying the same harmonic-proximity test to hemispherical monument corpora.

Null B (same-footprint restricted draw, 100,000 trials) is the discriminating comparison: does the dome/stupa subset show greater harmonic proximity than the broader UNESCO background occupying the same longitude footprint? This comparison is possible for the UNESCO corpus because a larger parent population (the full UNESCO Cultural/Mixed list,  $N = 1011$  sites) exists from which the dome/stupa subset was selected. No equivalent parent population exists for the Wikidata or OSM corpora, which are already operationally defined as all stupa-tagged items; a same-footprint Null B comparison therefore applies only to the UNESCO corpus. Three window widths were tested ( $\pm 2^\circ$ ,  $\pm 5^\circ$ ,  $\pm 10^\circ$ ; Bonferroni  $\alpha_{\text{adj}} = 0.0167$ ): the  $\pm 2^\circ$  window gives  $p = 0.0180$ ; the  $\pm 5^\circ$  window gives  $p = 0.0129$ ; the  $\pm 10^\circ$  window gives  $p = 0.0139$ . All three windows survive Bonferroni correction ( $\alpha_{\text{adj}} = 0.0167$ ), illustrating subset-specific enrichment within the shared geography across all window choices.<sup>2</sup>

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<sup>2</sup>A within-dome bootstrap (Null A, 100,000 trials) resamples from the dome sites' own longitude distribution; its expected hit rate is defined by the same geography carrying the proximity enrichment, so it cannot serve as a discriminating comparison. The observed A++ count is non-significant under this null ( $p = 0.5493$ , ns), confirming the pattern is localized to dome-site geography rather than an artifact of the comparison procedure.

## 5.4 Anchor Concordance

All three corpora recover the same  $\sim 35.25^\circ\text{E}$  Levantine corridor phase. The Normalized Phasor Consensus statistic  $K$  is defined as  $K = |\bar{u}_A \cdot \bar{u}_B|$ , where  $\bar{u}_c = (\sum_k e^{i\theta_k^{(c)}}) / |\sum_k e^{i\theta_k^{(c)}}|$  (unit complex numbers as in Section 2) is the normalized mean phasor of corpus  $c$  at the candidate period, and  $|\cdot|$  denotes the modulus of the dot product of two unit complex numbers. The dot-product formulation is preferred over the circular correlation coefficient (Mardia & Jupp, 2000, §8.3) because it compares phase *directions* only, stripping out amplitude information that is irrelevant when corpora differ substantially in size and clustering:  $K = 1$  when the two corpora point to the same phase,  $K = 0$  when they are orthogonal. Its null distribution is estimated by permutation throughout. In the worked example,  $K \geq 0.9675$  for all pairings of the three monument corpora (Table 4). The three-corpus combination reaches perm- $p = 0.02437$  with consensus anchor at  $0.029^\circ$  from  $35^\circ\text{E}$ . The primary non-nested two-corpus pair (Dome + Wikidata) is individually non-significant, reflecting the limited power of a two-corpus permutation test: high  $K$  indicates strong phase agreement, but two corpora alone cannot reject the null that such agreement arises by chance. Significance emerges when the independently assembled OSM corpus is added as a third data source, because three corpora independently converging on the same phase is rare under the permutation null even when individual pairs are underpowered. Negative controls show  $K$  drops sharply when the monument corpora are replaced with geographic or route-network controls.

Table 4: Selected NPC anchor-concordance configurations at  $T = 3^\circ$  (100,000 permutation draws). High  $K$  with non-significant two-corpus perm- $p$  reflects limited statistical power with only two corpora; three independently assembled corpora converging on the same phase is rare under the permutation null. The “Conservative” row pairs the broadest possible corpus (full UNESCO Cultural/Mixed) with the Wikidata stupa corpus, maximizing denominator size at the cost of corpus specificity. Negative controls (the final three rows) indicate  $K$  is not a generic Eurasian property.

| Role            | Corpora                        | $K$    | perm- $p$ | Note                    |
|-----------------|--------------------------------|--------|-----------|-------------------------|
| Non-nested pair | Dome + Wikidata Stupa          | 0.9675 | 0.16287   | Primary non-nested      |
| Conservative    | UNESCO full + Stupa            | 0.982  | 0.12243   | Broadest corpus pair    |
| Three-corpus    | Dome + Stupa + OSM             | 0.972  | 0.02437   | All 3 monument corpora  |
| Neg. control    | Americas + Wikidata Stupa      | 0.6723 | 0.29819   | Geographic control      |
| Contrast corpus | UNESCO + OWTRAD                | 0.7429 | 0.46532   | Route network           |
| Type control    | N. American mounds ( $N=236$ ) | —      | —         | No proximity enrichment |

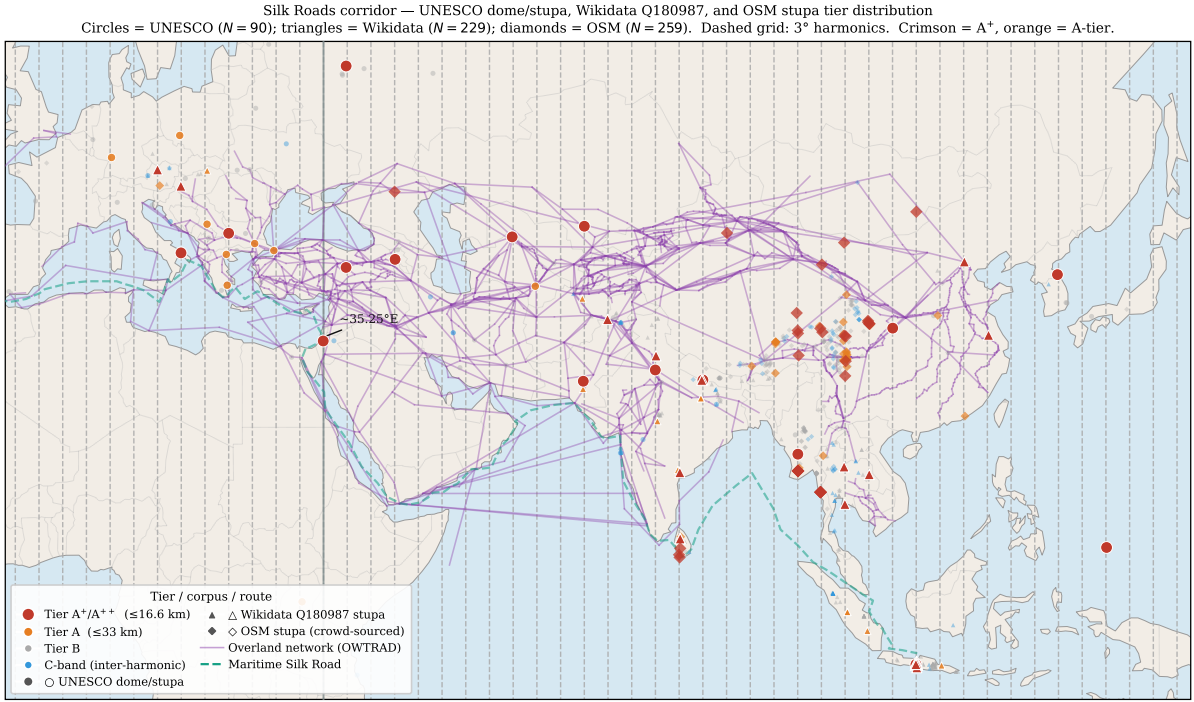


Figure 2: Silk Roads corridor showing all three monument corpora mapped against the  $3^\circ$  harmonic grid (dashed lines) anchored to the  $\sim 35.25^\circ\text{E}$  Levantine corridor phase. Circles = UNESCO dome/stupa ( $N = 90$ , keyword-matched); triangles = Wikidata Q180987 stupa ( $N = 229$ ); diamonds = OSM stupa ( $N = 259$ , crowd-sourced). Color indicates harmonic proximity tier: crimson = Tier A<sup>+</sup> or better ( $\leq 16.6$  km), orange = Tier A ( $\leq 33$  km), gray/blue = B/C-band. The dark meridian marks the fixed  $\sim 35.25^\circ\text{E}$  corridor phase. All results are conditional on the fixed corridor-phase frame. Map projection: equirectangular (geographic coordinates, WGS 84). Base map: Natural Earth (public domain, <https://www.naturalearthdata.com/>, accessed 1 March 2025).

## 6 Discussion

### 6.1 The Methods Contribution

The central methodological result is the separation between circular phase concentration and thresholded harmonic proximity. Rayleigh  $R$  describes whether the full phase distribution has a dominant mean direction at period  $T$ ; the geometric-null binomial describes whether a corpus contains more sites within a stated operational distance of any harmonic than expected from the fraction of longitude covered by those bands. Because the two statistics weight off-harmonic observations differently, they provide complementary information and can dissociate within the same corpus. This makes Rayleigh a useful diagnostic for phase structure and the geometric-null binomial the matched statistic for thresholded proximity-to-harmonics claims. In Silva’s terms, the tests detect and describe structured features in the spatial data; archaeological, historical, or geographic explanations for those features remain interpretive questions to be evaluated with independent evidence.

The analysis is a descriptive spatial test of harmonic proximity. Geographic clustering is part of the documented corpus pattern, so site-level binomial  $p$ -values describe corpus-level proximity enrichment rather than independent cultural decisions. The geometric null  $p_0 = 2d/T$  gives a transparent expected hit fraction under uniform longitude, and Fisher’s method is used as a cross-corpus consistency summary. This framework is applicable to any spatial periodicity hypothesis of this form regardless of geographic or cultural context.

### 6.2 Interpretation of the Empirical Illustration

The three-corpus worked example finds consistent directional excess over the geometric null at the fixed  $3^\circ$  period and  $\sim 35.25^\circ\text{E}$  Levantine corridor phase. The dome-excluded two-corpus combination (Wikidata + OSM) yields a Fisher combination of  $\chi^2(4) = 28.74$ ,  $p < 0.001$  at Tier A; the all-three-corpus result (Fisher  $p < 0.001$  at Tier A) includes the dome corpus and provides a less conservative cross-corpus summary than the dome-excluded combination. All results are frame-conditional proximity-enrichment measures

(see Limitations). The UNESCO corpus is the most geographically distributed result, with A-tier sites spanning the full Eurasian corridor.

Null B (same-footprint restricted draw) provides the discriminating comparison: among all UNESCO cultural sites occupying the same longitude footprint, the dome/stupa subset shows greater harmonic proximity than the broader background across all three window widths after Bonferroni correction (see Results), illustrating subset-specific enrichment within the shared geography. The cross-corpus phase concordance (NPC  $K \geq 0.9675$  across all pairings of three independently assembled monument datasets with different geographic centers of mass) is consistent with the phase not being an artifact of any single corpus’s geographic distribution: UNESCO dome sites, Wikidata Q180987 stupas, and OSM stupa tags converge on the same  $\sim 35.25^\circ\text{E}$  phase independently of one another.

**Corpus overlap and curation.** The three corpora are assembled by distinct processes: the UNESCO dome/stupa subset derives from keyword matching against the WHC XML dataset; Wikidata Q180987 was extracted via SPARQL query on the *instance-of: stupa* property; OSM was assembled by crowd-sourced tagging of **building=stupa**, **historic=stupa**, and **ruins=stupa** features via Overpass API. An empirical overlap check using GPS proximity (threshold  $< 1$  km) and name-string matching finds 4 near-coincident site pairs between UNESCO and Wikidata, 0 between UNESCO and OSM, and 1 between Wikidata and OSM (see supplementary overlap audit). Because all three datasets are publicly available, the candidate frame is handled as fixed across the empirical demonstration. The Fisher combination is used as a consistency check across differently assembled datasets. The geographic clusters of the three corpora are substantially non-overlapping at the region level: UNESCO dome sites span the full Eurasian corridor from the Levant to Java; Wikidata is dominated by the Java/Sumatra node; OSM coverage reflects crowd-sourced mapping density across South and Southeast Asia. No single shared geographic cluster explains enrichment in all three corpora simultaneously.

The Wikidata result reflects the documented geographic concentration of stupa con-

struction at the Java/Borobudur node (Miksic, 1990): that cluster’s position on a  $3^\circ$  harmonic is precisely the proximity structure the geometric-null test is designed to detect, and the geographic concentration is part of the documented corpus pattern rather than a confound to be removed. Removing the Java/Sumatra cluster would require applying the same logic to every regional node — the Levant, Sri Lanka, mainland Southeast Asia — at which point no corpus survives; the same arbitrary-threshold problem that motivates retaining all clusters (Section 3) applies here. The relevant check is cross-corpus: the corpus overlap audit finds only one near-coincident Wikidata–OSM pair, so the OSM enrichment is not a restatement of the same Borobudur sites under a different label. The OSM result, assembled by a process with no relationship to UNESCO or Wikidata curation, shows the same directional excess ( $p = 0.0059$  at Tier A) and the same phase concordance with Wikidata ( $K = 0.9695$ ) in the worked example. Figure 2 shows all three corpora mapped against the  $3^\circ$  harmonic grid.

The interpretation follows Silva (2020)’s distinction between statistical structure and archaeological explanation. Significance tests evaluate evidence against a null model; archaeological explanation is developed from evidence independent of the orientations themselves, including the wider archaeological record and socio-historical context. The statistical pattern illustrated here is compatible with several mechanisms. These include geographic channeling by trade routes and river systems following the Eurasian corridor, cultural diffusion patterns concentrating stupa construction at historically dominant nodes (Sen, 2014), and selection effects in the UNESCO inscription process. An exploratory positive finding is consistent with this: weighting OWTRAD nodes by degree centrality (the number of route edges incident to each node, as a proxy for hub importance) finds that the most network-central cities fall near the  $3^\circ$  harmonic grid more often than the uniform null predicts. If major route hubs and hemispherical monuments are both concentrated along the same interior Eurasian corridor, the harmonic proximity of each would be a shared consequence of corridor geography rather than an independent proximity result; disentangling these requires a geographic control outside the corridor, which remains a target for future work. Historical interpretations involving deliberate

placement would additionally require evidence for pre-modern continental longitude de-  
 termination at sub-degree precision (Hunger & Steele, 2019; Neugebauer, 1975). The  
 contribution of the present work is to establish what the directly matched statistical test  
 reports when applied to three differently assembled corpora.

## 6.3 Limitations

**Frame conditionality.** All  $p$ -values in Table 2 are conditional on the fixed  $3^\circ$  period  
 and  $\sim 35.25^\circ\text{E}$  Levantine corridor phase (anchored to Mount Gerizim,  $35.269^\circ\text{E}$ ). All  
 three corpora are evaluated against this same frame. The results are therefore corpus-  
 level proximity-enrichment measures under that fixed frame and should not be read as  
 outcomes of an unconditional search over all possible periods and anchors.

**Analysis sequence and exploratory scope.** This paper is a methods contribution  
 in the tradition of exploratory observational spatial analysis following Silva (2020): the  
 statistics describe spatial structure in the data and do not constitute evidence of his-  
 torical placement without independent archaeological support. The  $3^\circ$  period emerged  
 as the dominant proximity peak in the dome corpus sweep; the cross-corpus analysis is  
 the methodological demonstration that the same statistical pattern appears consistently  
 across three independently assembled datasets. In exploratory descriptive analysis this  
 is the appropriate sequence: identify a statistical pattern, then assess whether it is con-  
 sistent across independent datasets assembled by different processes. Readers should  
 interpret the Wikidata+OSM Fisher combination as the primary cross-corpus consis-  
 tency check; the three-corpus result, which includes the period-motivating corpus, is a  
 less conservative summary.

**Anchor-sweep correction.** The frame-conditional tests (Table 2) ask whether prox-  
 imity enrichment is present across corpora given the fixed frame. The anchor-sweep  
 correction asks a harder question: is this anchor globally exceptional after correcting for  
 the best of 36,000 trial longitudes? The corrected  $p = 0.0753$  ( $z = 1.65$ , marginal) quan-  
 tifies what the evidence would look like under an unconstrained global search. That the  
 global correction is marginal while the frame-conditional evidence is strong is not a con-

tradiction: the max-perm correction accounts for the full cost of an unconstrained search, which the frame-conditional analysis does not require. The frame-conditional results are well-supported within the fixed frame; the anchor should not be treated as the outcome of a globally optimal search.

## 7 Conclusion

This paper demonstrates the distinction between Rayleigh concentration and thresholded harmonic proximity as descriptors of a modulo- $T$  phase distribution: Rayleigh summarizes mean circular concentration, while the geometric-null binomial tests thresholded proximity to any harmonic under the benchmark  $p_0 = 2d/T$ . The period sweep illustrates this distinction empirically, with Rayleigh-significant periods lacking proximity enrichment and the enriched  $3.0^\circ$  period lacking significant Rayleigh concentration. Applied under a fixed operational frame — fixed period, corridor phase, thresholds, and corpora — the binomial framework finds a consistent corpus-level proximity pattern across three hemispherical-monument datasets at the  $3^\circ$  period and  $\sim 35.25^\circ$  E Levantine corridor phase.

Two qualifications contextualize these results, both detailed in the Limitations section. The period emerged from the dome corpus sweep; the Wikidata+OSM combination ( $\chi^2(4) = 28.74$ ,  $p < 0.001$  at Tier A) is the primary cross-corpus check precisely because it excludes that corpus. The anchor-sweep-corrected  $p = 0.0753$  quantifies what the evidence looks like under an unconstrained global search — a harder question than the frame-conditional one, answered separately.

Following [Silva \(2020\)](#), the worked example is best read as a descriptive spatial exercise. This paper is a methods contribution: it introduces and validates the geometric-null binomial framework, and uses three hemispherical-monument corpora as a worked example to illustrate how the method behaves on real data. No confirmatory historical judgement is made, and none is possible from these results alone. The statistics report corpus-level proximity enrichment under a fixed frame; they cannot identify causal or

intentional relationships. This is an open-data observational analysis: the datasets are publicly available and every step is independently reproducible from the published code and audit files. The evidentiary gap between “sites are disproportionately close to a 3° harmonic grid” and “sites were placed with knowledge of that grid” lies outside the scope of any purely statistical test; such claims require independent archaeological and socio-historical evidence that proximity statistics cannot supply.

The paper’s contribution is the explicit demonstration that Rayleigh concentration and geometric-null binomial proximity can dissociate within the same corpus — with each statistic remaining useful for its own question — and a reproducible open-data example that future comparative work can evaluate.

All data, code, and audit files are openly available at Zenodo (Tenenbaum 2026; <https://doi.org/10.5281/zenodo.19938283>).

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## 530 Data Availability

531 All data, code, and audit files: Zenodo (Tenenbaum 2026; [https://doi.org/10.5281/](https://doi.org/10.5281/zenodo.19938283)  
532 [zenodo.19938283](https://doi.org/10.5281/zenodo.19938283), confirmed resolving May 2026). UNESCO XML (UNESCO World  
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544 **Seth Tenenbaum:** Conceptualization, Methodology, Software, Formal analysis, Data  
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## 546 Declaration of Generative AI and AI-Assisted Technolo- 547 gies in the Writing Process

548 During the preparation of this work the author used Claude (Anthropic) to assist with  
549 prose drafting and GitHub Copilot (Microsoft) to assist with code autocompletion. After  
550 using these tools, the author reviewed and edited all content as necessary and takes full  
551 responsibility for the content of the published article.